# Statement of Research Accomplishments and Future Plans

I am a computational plasma physicist with expertise on both the Magnetohydrodynamic (MHD) and particle-in-cell (PIC) simulations of space plasma processes. My research lies at the intersection of magnetic reconnection, and turbulence in space and astrophysical plasmas. Magnetic reconnection refers to a process where the magnetic field topology is re-arranged and magnetic energy is converted to particle energy while turbulence, on the other hand, is the chaotic behavior of plasma.

In my Ph.D., I have investigated *the interplay between magnetic reconnection and turbulence* using two-dimensional particle-in-cell (PIC) simulations. This work primarily focused on the spectral and dynamical characteristics of energy transfer in laminar (vanilla) reconnection and cross-compared them with that of decaying plasma turbulence. The primary findings include:

1. *Kolmogorov Spectra in Reconnection:* Laminar reconnection with no initial seed of turbulence generates a Kolmogorov like magnetic energy spectrum with a -5/3 slope at sub-ion-inertial length scales. This spectral feature is identical to that in turbulence [1].
2. *Energy Cascade Across Scales:* Energy transfer across the intermediate scales in reconnection is a cascade, a trait observed in turbulence [2].
3. *Influence of Guide Fields:* The turbulence-like features of magnetic reconnection become more pronounced with an external guide magnetic field, strengthening the observed correlation between reconnection rates and turbulent energy transport [3].
4. *Dissipation via Pressure-Strain Interaction:* Pressure strain interaction is a pathway of energy transfer between thermal and fluid flow energy. A coarse-grained analysis reveals that pressure-strain interaction mediates energy conversion at kinetic scales and serves as a robust proxy for energy dissipation in both reconnection and turbulence [4].

These findings have advanced on the fundamental understanding of the interrelationship between reconnection and turbulence, with implications for understanding space plasma heating. It has also opened a new avenue of space plasma research.

## *Plans forward: Future Research*

I see an exciting opportunity to establish a robust space plasma physics program at the University of Wyoming, especially through collaboration with NWSC and leveraging high-performance computing resources. My immediate and long-term research agenda focusses on space plasma turbulence. These include finding answers to the following research questions

1. *How does magnetic, thermal pressure, and their relationship influence turbulent structures in the heliosphere?*
2. *How does magnetic and flow shear affect each other in space plasma processes such as magnetic reconnection and turbulence?*
3. *How is energy transferred differently in reconnection events involving just electrons versus those involving heavier ions?*

To address these questions, some of the immediate research projects include:

1. *Pressure Balance in Kinetic Plasmas:* Statistical studies of pressure fluctuations in kinetic plasmas have shown that the well-known anti-correlation between thermal and magnetic pressures seen in MHD turbulence also holds in kinetic regimes [5], especially at scales near the ion scales. This suggests that total pressure (sum of magnetic and thermal) balance persists across plasma models, but further investigation is required. Some of the immediate extension to this work include:

- Examining pressure balance structures in 3D kinetic simulations of decaying turbulence.
- Explore the role of pressure anisotropy (variation along the perpendicular and parallel plane) in pressure-balance structures using both 2D MHD and PIC models.
- Investigate the implications of pressure balance structures on turbulent stresses and energy transport

2. *Shear Flow in Electron-Only Reconnection:* Electron-only reconnection [6] is a reconnection where only electrons participate and occurs at electron scales. This type of reconnection is recognized as a key mechanism in energy dissipation in the solar wind. However, the influence of shear in particle flow, ubiquitous in magnetospheric and laboratory plasmas on the electron-only reconnection remains largely unexplored. Using kinetic PIC simulations, I plan to:

- Systematically vary the amplitude of the electron shear flow and examine its effect on the strength of reconnection, electron heating, and magnetic flux transport.
- Clarify the role of electron shear flow in energy conversion in the magnetosheath and dayside magnetopause, natural sites for electron-only reconnection.

3. *Magnetic Shear in Kelvin-Helmholtz Instability (KHI):* Kelvin-Helmholtz instability (KHI) originates due to shear flow and is as common driver of plasma turbulence at shear interfaces. While previous works has examined small-scale dissipation mechanisms [7], the role of magnetic shear remains unresolved. This project will involve:

- Performing PIC simulations of KHI with varying degrees of magnetic shear.
- Investigating how magnetic shear modifies nonlinearity, coherent structures, and turbulent energy spectra in shear turbulence.
- Contributing to a broader understanding of shear flow turbulence in both space and laboratory environments in relation to project 2 (See above).

I have secured *20,000 node hours* at the DOE's NERSC, a supercomputing facility to conduct a parametric study quantifying magnetic shear effects in shear-driven turbulence.

4. *Pressure-Strain Interaction in Magnetic Reconnection and Turbulence:* Pressure-strain interaction plays a pivotal role in energy conversion beyond the ion scales. In a collaboration with researchers at the University of Delaware (Delaware), Clemson University (South Carolina), Laboratory for Atmospheric and Space Physics (Colorado), and European institutions CNRS (France) to:

- Map the evolution of pressure-strain interaction from ion-coupled (where ion also take part in magnetic reconnection) to electron-only reconnection regimes.
- Analyze how parameters such as guide field strength, initial temperature, and current sheet width affect energy conversion.
- Extend this analysis to turbulent systems, exploring how turbulence amplitude and spectral content shape pressure-strain behavior.

In addition, I have recently started working with the observational data from the Parker Solar Probe (PSP), spacecraft closest to the sun. An immediate project on the observational aspect include

5. *Parker Solar Probe (PSP) Observations:* Analysis of the magnetic field data from PSP encounters to investigate sub and super-Alfvénic (based on the Mach number) turbulence in the near-Sun environment. In collaboration with Mahidol University (Thailand) and the University of Delaware (Delaware), I plan to:

- Identify turbulence regimes and compare with kinetic simulation predictions
- Bridge the gap between in-situ observations and kinetic-scale numerical modeling

Overall, these projects will help us understand magnetic reconnection and plasma turbulence in the space environments.

## *Vision for Space Physics at the University of Wyoming*

My research combines simulations, theory, and real spacecraft data, aligning well with the University of Wyoming's strengths in computational science. I bring ongoing collaborations with national and international partners and plan to actively involve students—especially through hands-on projects and mentorship opportunities.

I am also deeply committed to education and mentorship. I plan to involve graduate and undergraduate students, particularly through small-scale computational research projects, including REU style experiences. While I don't currently have dedicated funding, I plan to apply for support through NASA, the NSF, and Wyoming-based research programs. Additional student funding mechanisms are also available through the Space Grant and NASA graduate research programs. These efforts will help build a strong pipeline for training students in this exciting and rapidly evolving field.

## *Conclusion*

I'm passionate about exploring the fundamental physics that shapes our space environment. With a strong foundation in plasma modeling, experience working with spacecraft data, and a commitment to collaborative science and mentorship, I am ready to launch a vibrant research program at the University of Wyoming. I look forward to helping train the next generation of scientists who will explore the mysteries of the Sun, solar wind, and beyond. If selected, I look forward to contributing to the department's mission and training the next generation of space physicists.

## *References*


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